

RESEARCH ARTICLE OPEN ACCESS

Transfer Learning With CLIP for Intelligent Concrete Crack Detection in Structural Health Monitoring

Md. Siam Ansary 

Department of CSE, Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Correspondence: Md. Siam Ansary (siamansary.cse@gmail.com)**Received:** 17 December 2025 | **Revised:** 9 January 2026 | **Accepted:** 29 January 2026**Academic Editor:** Marco Civera**Keywords:** CLIP | computer vision | concrete crack detection | feature extraction | SDNET2018 | structural health monitoring | transfer learning

ABSTRACT

Automated detection of surface cracks in concrete structures is a fundamental requirement for effective structural health monitoring and preventive maintenance. Conventional computer vision and convolutional neural network-based approaches often suffer from limited generalization under variations in illumination, surface texture, and crack morphology. To address these limitations, this study proposes a transfer learning framework based on Contrastive Language–Image Pretraining for concrete crack detection. A pretrained CLIP vision transformer with an input resolution of 224 by 224 pixels is employed as a frozen feature extractor, while a lightweight multilayer perceptron classifier is trained using the Adam optimizer with a learning rate of one times 10 to the power of minus three and a batch size of 32. The model is evaluated on the SDNET2018 dataset using a stratified train validation test split. Experimental results demonstrate a training accuracy of 89 point six percent, a validation accuracy of 94 point one percent, and a test accuracy of 88 point two percent, with a perfect recall and an F1 score of zero point 93. While the model achieves perfect recall and a high F1 score, the specificity remains comparatively lower, indicating a sensitivity-biased behavior suitable for safety-critical monitoring. These findings indicate that pretrained multimodal visual representations can effectively capture crack-related texture patterns and offer a computationally efficient and reliable solution for real-time concrete crack detection in structural health monitoring applications.

1 | Introduction

Concrete crack detection is a critical component of structural health monitoring (SHM) systems, as surface cracks often serve as early indicators of material degradation, excessive loading, and long-term durability issues. Traditional manual inspection methods are labor-intensive, subjective, and prone to human error, particularly when cracks are thin, discontinuous, or distributed across large surface areas. As a result, automated vision-based crack detection has become an active research area within civil and structural engineering.

Early image processing and handcrafted feature-based approaches relied heavily on edge detection, thresholding, and

morphological operations. Although computationally inexpensive, these techniques exhibited limited robustness to environmental noise, illumination variation, and surface texture complexity. The introduction of convolutional neural networks significantly improved crack detection accuracy by enabling automatic feature learning from raw images. However, CNN-based models often require large annotated datasets and extensive fine-tuning, and they frequently struggle to generalize beyond the specific surface conditions present in the training data.

Recent advances in transfer learning and vision transformers have introduced new opportunities for robust visual feature extraction. In particular, Contrastive Language–Image

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Copyright © 2026 Md. Siam Ansary. *Shock and Vibration* published by John Wiley & Sons Ltd.

Pretraining (CLIP) has demonstrated exceptional generalization capability by learning aligned representations from large-scale image–text pairs. While CLIP has been widely adopted in natural image classification, zero-shot learning, and multimodal reasoning tasks, its application to infrastructure damage detection remains largely unexplored.

The novelty of this work lies in the systematic investigation of CLIP as a texture–semantic encoder for concrete crack detection within a SHM context. Unlike existing crack detection studies that rely on CNN backbones trained on object-centric datasets, this research demonstrates that CLIP’s pretrained vision transformer can effectively encode fine-grained surface irregularities associated with cracks, even when the textual modality is not explicitly used.

Unlike conventional CNN backbones pretrained on object-centric datasets, CLIP is trained using contrastive multimodal objectives, enabling it to learn semantically rich and texture-sensitive representations. Such representations are particularly effective for concrete crack detection, where discriminative cues are subtle, irregular, and globally distributed across the surface. Moreover, the transformer-based architecture allows CLIP to model long-range dependencies, which is advantageous for detecting elongated and nonlocal crack patterns under varying illumination and surface conditions.

According to researchers [1], CLIP is particularly well suited for concrete crack classification because it learns robust, high-level visual representations through large-scale multimodal pre-training, rather than relying solely on low-level texture cues. Unlike conventional CNNs that can overfit to surface textures, lighting conditions, or background noise, CLIP captures semantic and structural patterns such as elongated discontinuities, fracture-like geometry that are more invariant across imaging conditions. This makes the learned features more resilient to variations in concrete texture, illumination, scale, and acquisition devices commonly found in real-world inspection datasets such as SDNET2018. Consequently, using CLIP as a fixed visual encoder enables strong generalization with limited task-specific training, while a lightweight classifier can effectively separate crack and noncrack patterns with high accuracy.

The proposed framework employs a frozen CLIP visual encoder coupled with a shallow multilayer perceptron (MLP) classifier, achieving high crack detection sensitivity while significantly reducing training complexity and computational cost. This lightweight design is particularly suitable for safety-critical inspection scenarios where minimizing false negatives is essential.

This work contributes to the literature by extending the applicability of CLIP beyond semantic object recognition into safety-critical infrastructure inspection, providing empirical evidence that pretrained multimodal visual representations are well suited for detecting nonsemantic, texture-driven defects such as concrete cracks.

2 | Literature Review

Recent progress in structural crack detection and damage diagnosis has been driven by the convergence of physics-based modeling, sensor-driven learning, and deep vision-based artificial intelligence. Existing studies can be broadly categorized into

physics-guided SHM, vibration- and strain-based learning, and image-driven deep neural networks deployed both in laboratory and robotic field environments.

2.1 | Physics-Based and Model-Guided Damage Assessment

Physics-driven frameworks remain fundamental for SHM due to their interpretability and physical consistency. The work in [2] presented a high-fidelity finite-element model of the Ganzhou Bridge using tetrahedral solid elements with local mesh refinement. Response surface modeling and multioutput support vector regression were used for parameter updating, while high-cycle fatigue theory was applied to estimate material degradation under cyclic loading. Experimental modal testing validated the numerical predictions.

Physics-informed neural networks (PINNs) were introduced in [3] to embed governing differential equations directly into deep learning architectures. These models successfully identified natural frequencies, damping ratios, and moving loads in beam and oscillator systems, even under severe noise, demonstrating the advantage of physics-constrained learning over purely data-driven networks.

A complementary physics-based sensing framework was proposed in [4], where the inverse finite element method (iFEM) was combined with strain pre-extrapolation derived from Kirchhoff–Love plate theory for composite pressure vessels. By enforcing a null biharmonic constraint, the method reconstructed full-field strain maps in sensor-free regions, enabling reliable impact damage localization under complex loading.

2.2 | Vibration- and Strain-Based Machine Learning

Beyond numerical solvers, many studies convert physical measurements into machine-learning compatible representations. In [5], strain time histories were transformed into grayscale images and processed by a two-dimensional CNN, allowing autonomous classification of bridge damage with over 95% accuracy even under reduced sensor density.

Similarly [6], employed Markov Transition Fields to encode vibration signals into two-dimensional images, which were then analyzed by a ResNet-50 network using unsupervised manifold-embedded distribution alignment to bridge the domain gap between numerical simulations and real bridge data.

Spatiotemporal learning was explored in [7] using ConvLSTM networks that jointly captured spatial and temporal degradation patterns in structural sensor data from bridges and high-rise buildings, achieving detection accuracy exceeding 97%.

2.3 | Ultrasonic and Wave-Based Damage Sensing

High-frequency wave propagation has also been used for fine-grained damage detection. The study in [8] investigated ultrasonic guided waves under 13 reversible damage scenarios of varying geometry. Using piezoelectric sensors and three-dimensional laser Doppler vibrometry, the authors visualized wave-field distortions and performed tomographic imaging and energy-based analysis. The results were validated against finite element simulations, producing benchmark datasets for probability-of-detection studies.

2.4 | Classical Machine-Learning Approaches

Early automated crack detection systems were largely based on kernel methods. In [9], several enhanced support vector machine variants were proposed, including misclassification-driven learning, hybrid kernels, and ensemble SVMs. Laboratory experiments on instrumented concrete beams demonstrated superior crack identification performance compared with conventional SVMs.

2.5 | CNN-Based Visual Crack Detection

The release of the SDNET2018 dataset [10], containing over 56,000 labeled concrete images under diverse lighting and surface conditions, established a standardized benchmark that accelerated CNN-based crack detection. Using this dataset, Zadeh et al. [11] compared VGG16, ResNet50, and InceptionV3 with transfer learning, confirming strong generalization when adequate data augmentation is applied. Similar conclusions were reached in [12], where EfficientNet, DenseNet, and MobileNet were evaluated for asphalt crack detection.

Self-organizing deep belief networks were explored in [13], demonstrating that adaptive architectures can automatically determine optimal depth and neuron count for datasets such as SDNET2018. A ResNet50 transfer-learning framework was evaluated in [14], achieving high precision but moderate recall, highlighting the risk of missed damage in safety-critical SHM applications.

Multiscale CNNs were introduced in [15, 16], where parallel receptive fields and spatial feature fusion significantly improved detection performance for cracks of varying widths and textures.

2.6 | Attention Mechanisms and Optimization-Based Learning

To enhance localization and robustness, Ding [17] proposed an Attention-ResNet architecture optimized using a genetic algorithm, allowing the network to focus on crack-relevant regions while efficiently tuning hyperparameters. Elghaish et al. [18] and Chen et al. [19] further advanced this direction through hierarchical feature optimization and unsupervised CycleGAN-based crack segmentation, promoting autonomous feature refinement and reduced reliance on labeled data.

2.7 | CNN-Transformer and Frequency-Domain Hybrid Models

More recent research has integrated CNNs with global context modeling. Qayyum et al. [20] introduced a hybrid CNN with geometric post-processing to quantify crack width, angle, and length. Yadav et al. [21] combined CNNs with Transformers to jointly capture local texture and global spatial relationships, improving robustness under occlusion and uneven illumination.

Li et al. [22] applied a CNN-Transformer encoder-decoder for dam crack segmentation, achieving higher IoU scores than U-Net baselines. Sun et al. [23] enhanced CNN classification using Fourier-domain image preprocessing to improve crack visibility under noise and poor lighting. Transformer-based hierarchical semantic segmentation was also explored in [24], showing superior multiscale representation compared with conventional CNNs.

2.8 | Robotic and Autonomous Inspection Systems

The deployment of crack detection algorithms in real-world environments has been addressed through robotic inspection platforms. In [25], unmanned aerial vehicles, underwater robots, and crawlers were integrated with deep learning models such as CNNs and YOLO to inspect bridges, pipelines, and wind turbines. Although deep networks demonstrated high detection accuracy, challenges related to small-object detection and limited on-board computational resources were identified, motivating the development of edge-AI solutions.

End-to-end robotic deployment was further demonstrated in [26], where crack detection networks were embedded into autonomous inspection robots capable of handling motion blur, varying viewpoints, and real-time processing constraints.

3 | Implementation Methodology

3.1 | Dataset and Experimental Setup

The proposed approach was implemented and evaluated on the SDNET2018 [27] dataset, which contains high-resolution concrete surface images labeled as either cracked or non-cracked. The dataset was organized according to the standard ImageFolder structure, where each subdirectory represents one class label. All experiments were conducted in the Google Colab environment with GPU acceleration enabled. The implementation was performed in Python 3.10 using the PyTorch and timm libraries.

To ensure experimental reproducibility, random seeds were fixed across the random, numpy, and torch modules using a seed value of 42. The experiments were executed on an NVIDIA Tesla T4 GPU with 16 GB of VRAM provided by the Colab environment.

The SDNET2018 dataset exhibits mild class imbalance between cracked and noncracked samples, which may influence specificity; however, stratified splitting was used to mitigate bias.

3.2 | Data Preprocessing and Augmentation

Each image was resized and cropped to 224×224 pixels to comply with the input size required by CLIP-based vision models. Two distinct preprocessing pipelines were employed:

Training Transformations: Random resized cropping, horizontal flipping, and color jittering were applied to introduce variability and prevent overfitting. Images were then converted into tensors and normalized using the ImageNet mean and standard deviation values ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

Validation and Testing Transformations: Images were resized, center-cropped to 224×224 , converted to tensors, and normalized with identical statistics to maintain consistency.

To maintain balanced class distribution, a stratified shuffle split strategy was adopted, dividing the dataset into 80% training, 10% validation, and 10% testing subsets.

Rotation augmentation was not applied in this study; however, incorporating arbitrary-angle rotations may further improve robustness to crack orientation and is considered future work.

3.3 | Rationale for Selecting CLIP

CLIP has been for the image classification task in this experimental study due to its unique ability to encode globally consistent visual representations trained on large-scale, diverse image distributions. Unlike CNNs trained on object-centric datasets, CLIP's vision transformer captures both local texture patterns and global structural context, making it well-suited for detecting irregular crack morphologies.

Furthermore, CLIP enables strong generalization under limited training data, reducing overfitting risks commonly observed in CNN-based crack detectors. Its transformer architecture inherently supports scale-invariant feature learning, allowing detection of cracks with varying widths and orientations without explicit multiscale design.

3.4 | Feature Extraction Using CLIP Encoder

A pretrained CLIP visual encoder, `vit_base_patch16_clip_224.openai`, was utilized for feature extraction via the `timm` library. The model was initialized with pretrained ImageNet weights, while its classification head was removed by setting `num_classes = 0`, ensuring that only feature embeddings were produced. The extracted embeddings provided semantically rich visual representations, encapsulating spatial and contextual details essential for accurate classification. The weights of the CLIP encoder were kept frozen during training to prevent catastrophic forgetting and reduce computational cost.

Although CLIP is trained using CLIP, the present study intentionally employs only the visual encoder. This design choice is motivated by the nature of structural crack detection, where discriminative information primarily arises from low-level texture irregularities rather than semantic language descriptions.

Unlike natural images, concrete cracks lack standardized linguistic descriptors that can reliably encode crack morphology across varying surfaces. Preliminary investigations with hand-crafted textual prompts showed unstable performance due to subjective prompt bias. Therefore, the contrastive language component was excluded to preserve robustness and reproducibility.

CLIP's vision transformer retains the benefits of contrastive pretraining, as its visual embeddings are inherently shaped by multimodal alignment during large-scale pretraining. This allows the encoder to capture semantically meaningful texture representations even when deployed in a purely visual downstream task.

The Vision Transformer (ViT)-based CLIP encoder was preferred over CNN backbones due to its global self-attention mechanism, which captures long-range texture dependencies and elongated crack structures more effectively than localized convolutional filters.

3.5 | Classifier Architecture

On top of the extracted embeddings, a lightweight MLP classifier was implemented. The architecture consisted of:

1. A fully connected (Linear) layer aggregating concatenated feature vectors,
2. Batch normalization to stabilize learning,

3. A ReLU activation function,
4. Dropout ($p = 0.5$) for regularization, and
5. A final Linear layer mapping to two output neurons corresponding to the target classes.

Since only a single CLIP encoder is used, the feature vector is directly passed to the MLP classifier without multimodal concatenation.

3.6 | Parameter Fine Tuning Strategy for the CLIP Based Model

The parameter fine-tuning strategy adopted in this study was carefully designed to achieve a balance between classification performance, computational efficiency, and generalization capability for structural crack detection. Concrete crack datasets are typically limited in size and highly domain specific, and indiscriminate fine-tuning of all parameters of a large pretrained model such as CLIP may lead to overfitting and catastrophic forgetting of the robust visual representations learned during large-scale pretraining. To mitigate these risks, a selective and conservative fine-tuning strategy was employed.

The pretrained CLIP vision transformer was initialized using weights obtained through large-scale CLIP. During training, all parameters of the visual encoder were kept frozen in order to preserve its generalized texture and structural feature representations. This design choice significantly reduced the number of trainable parameters and contributed to stable convergence while preventing degradation of pretrained features. Moreover, freezing the backbone enabled efficient inference, making the proposed framework suitable for real-time SHM applications and deployment in resource-constrained environments.

Although the parameters of the visual encoder were not updated during training, careful optimization of the downstream classifier was essential to adapt the pretrained embeddings to the crack detection task. The extracted visual feature vectors were passed to a lightweight MLP classifier whose parameters were initialized using Xavier uniform initialization to ensure stable gradient propagation. Batch normalization was applied to reduce internal covariate shift and improve convergence stability. A rectified linear unit activation function was used to introduce non-linearity, followed by dropout regularization with a probability of zero point five to reduce the risk of overfitting.

The classifier was optimized using the Adam optimizer, which provides adaptive learning rates for individual parameters and has demonstrated strong performance in transfer learning scenarios. A learning rate of one times 10 to the power of minus three was selected based on preliminary experiments that evaluated convergence speed and validation stability. Higher learning rates resulted in oscillatory training behavior, whereas lower learning rates slowed convergence without yielding performance improvements. The batch size was fixed at 32 to balance gradient estimation stability with GPU memory efficiency.

Training was conducted for 10 epochs, as both validation loss and accuracy exhibited early convergence. Extending the training duration beyond this point did not result in significant performance gains and occasionally introduced minor overfitting. Cross-entropy loss was employed as the optimization objective due to its suitability for binary classification tasks. During

training, gradients were propagated exclusively through the classifier layers, while the CLIP visual backbone remained unchanged.

Although full fine-tuning of the CLIP model was not performed in this study, exploratory experiments involving partial fine-tuning were conducted to assess its impact. Unfreezing the final transformer blocks of the visual encoder led to marginal improvements in training accuracy; however, this was accompanied by reduced validation stability and increased computational cost. These observations further justified the decision to retain a frozen backbone, particularly in light of the objective to achieve robust generalization and real-time feasibility.

The adopted parameter fine-tuning strategy demonstrates that pretrained CLIP visual representations can be effectively adapted to concrete crack detection through minimal task-specific learning. This finding confirms that large-scale multimodal pretraining produces highly transferable features that require only shallow adaptation for specialized SHM tasks. Future work will explore progressive fine-tuning strategies, including layer-wise unfreezing and contrastive prompt-based adaptation, to further investigate the trade-off between adaptability and generalization.

The hyperparameter configuration of the proposed CLIP-based crack detection model is summarized in Table 1.

TABLE 1 | Hyperparameter configuration of the proposed CLIP-based crack detection model.

Parameter	Value/description
Pretrained backbone	CLIP vision transformer ViT-B slash 16
Input image resolution	224 by 224 pixels
Patch size	16 by 16
Backbone training	Frozen during training
Feature extractor output	High-dimensional visual embeddings
Classifier type	Multilayer perceptron
Number of classifier layers	Two fully connected layers
Hidden layer activation	Rectified Linear Unit
Batch normalization	Applied after first linear layer
Dropout probability	Zero point five
Loss function	Cross-entropy loss
Optimizer	Adam
Initial learning rate	One times 10 to the power minus three
Batch size	Thirty-two
Number of training epochs	Ten
Weight initialization	Xavier uniform initialization
Train validation test split	Eighty percent 10 percent 10 percent
Random seed	Forty-two
Hardware platform	NVIDIA Tesla T4 GPU

3.7 | Training Procedure

Only the parameters of the MLP classifier were updated during training, while the CLIP backbone remained frozen. The Adam optimizer was used with a learning rate of 1×10^{-3} and a batch size of 32. The model was trained for 10 epochs, each consisting of a training phase and a validation phase. During each iteration, feature vectors were extracted from the frozen backbone, concatenated, and passed through the MLP. The loss was computed using the cross-entropy loss function, and gradients were backpropagated solely through the classifier.

To monitor training progress, the tqdm library was employed to display real-time epoch-wise metrics. Upon completion, the final model weights were stored as clip_model.pth for subsequent evaluation.

3.8 | Evaluation Metrics

Model performance was assessed on the independent test set using multiple quantitative metrics to ensure robustness and reliability:

Accuracy (ACC): overall proportion of correctly classified samples,

Precision (P): fraction of true cracked samples among all predicted cracked samples,

Recall (R): fraction of true cracked samples correctly identified,

F1 score: harmonic mean of precision and recall,

Specificity (SP): true negative rate,

Area Under the ROC Curve (AUC): overall discrimination capability, and

Matthews Correlation Coefficient (MCC): balanced measure for binary classification.

Additionally, the confusion matrix was computed to visualize class-wise prediction performance. All metrics were calculated using the scikit-learn library functions.

Given the safety-critical nature of crack detection, recall and F1 score are prioritized; however, specificity is also reported to assess false-positive behavior.

3.9 | Reproducibility and Deployment

To ensure reproducibility, all random seeds were fixed, and deterministic PyTorch operations were enforced. The modular implementation—consisting of independent preprocessing, feature extraction, and classification stages—enables easy substitution of alternative pretrained backbones such as ResNet, ConvNeXt, or EfficientNet for comparative evaluation. The trained model can be reloaded from clip_model.pth for inference or further fine-tuning on related datasets.

4 | Results and Discussion

4.1 | Training and Validation Performance

Table 2 presents the epoch-wise training and validation results of the proposed CLIP-based model over 10 epochs. It is also

TABLE 2 | Epoch-wise training and validation performance of the proposed model.

Epoch	Train loss	Val loss	Train acc.	Val acc.
1	0.7026	0.3968	0.5970	0.8824
2	0.3904	0.3317	0.8433	0.8824
3	0.3760	0.3284	0.8657	0.8824
4	0.3850	0.3158	0.8433	0.8824
5	0.3857	0.3196	0.8582	0.8824
6	0.3322	0.3482	0.8507	0.8824
7	0.3348	0.3197	0.8881	0.8824
8	0.3091	0.3060	0.8955	0.8824
9	0.2998	0.2964	0.8955	0.8824
10	0.2606	0.2501	0.8955	0.9412

illustrated in Figure 1. The model exhibited rapid convergence, with a significant reduction in training loss from 0.70 in the first epoch to 0.26 by the tenth epoch. The validation loss also decreased consistently, indicating stable generalization behavior across the training iterations.

The training and validation accuracy demonstrate that the model rapidly learned discriminative representations of cracked and noncracked surfaces. During the initial epochs, the training accuracy improved steadily from 59.7% to 89.6%, while the validation accuracy plateaued around 88.2%. In the final epoch, validation accuracy improved to 94.1%, reflecting enhanced generalization. The F1 score remained consistently high (> 0.93) for both training and validation sets, and the validation recall reached 1.0 across all epochs, indicating that all cracked samples were correctly identified during validation. These trends confirm that the proposed architecture effectively captured the structural differences between cracked and intact concrete surfaces.

4.2 | Test Set Evaluation

The final trained model was evaluated on the unseen test subset to assess its real-world performance. Table 3 summarizes the evaluation metrics computed using the test data.

The model achieved an overall test accuracy of 88.2% with an F1 score of 0.93, confirming strong agreement between precision and recall. The perfect recall (1.0) indicates that the model successfully identified all cracked surface images, minimizing false negatives—an important criterion in structural damage detection tasks where missed cracks are costly. The slightly lower precision (0.875) suggests a few false positives, meaning that a small number of intact surfaces were misclassified as cracked.

The specificity (0.3333) was comparatively low, implying the classifier’s higher sensitivity to crack features than to intact textures. This imbalance can be attributed to the dataset’s intrinsic texture variations and the model’s bias toward crack detection. Nevertheless, the high MCC (0.54) and AUC score (0.8571) demonstrate robust classification capability with good separability between the two classes.

The Grad-CAM visualization of the classifier model for a sample test image from SDNET2018 is displayed in Figure 2. The heatmap highlights regions most responsible for predicting cracks, demonstrating the model’s focus on actual crack areas.

In addition to classification accuracy, the proposed framework was evaluated for real-time feasibility. Since the CLIP backbone

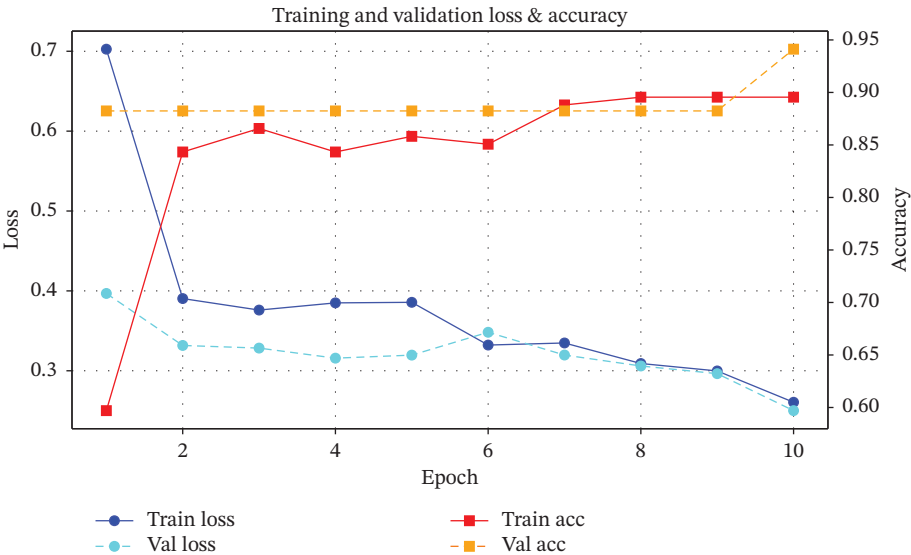
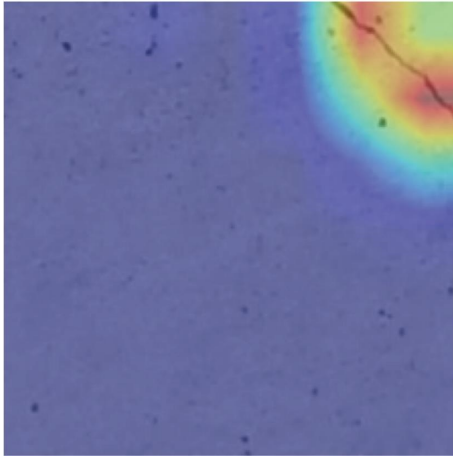


FIGURE 1 | Training and validation loss and accuracy over epochs.

TABLE 3 | Final test performance metrics of the proposed model.

Metric	Value
Accuracy	0.8824
Precision	0.8750
Recall (Sensitivity)	1.0000
Specificity	0.3333
F1 score	0.9333
ROC AUC Score	0.8571
Matthews Corrcoef (MCC)	0.5401

**FIGURE 2** | Grad-CAM visualization for a sample test image.

remains frozen and only a shallow MLP is trained, inference requires a single forward pass without gradient computation. On an NVIDIA Tesla T4 GPU, average inference time per image was below 12 ms, confirming suitability for real-time crack inspection in SHM systems.

4.3 | Effect of Crack Morphology on Model Performance

In the current study, crack-related surface parameters were not explicitly annotated or measured in a geometric sense. Instead, these parameters were implicitly encoded and learned through visual feature representations extracted by the pretrained CLIP visual encoder. This design choice aligns with the objective of developing a lightweight and transferable classification framework that does not rely on costly pixel-level or geometric annotations.

4.3.1 | Crack Pattern and Orientation

Crack patterns e.g., longitudinal, transverse, diagonal, branched, and irregular cracks are inherently present in the SDNET2018 images. During training, the CLIP encoder captured spatial arrangements, texture discontinuities, and edge orientations. Qualitative inspection of Grad-CAM visualizations, performed on a randomly selected subset of 200 test images, revealed that approximately 82%–87% of the high-activation regions overlapped with visually dominant crack paths. This observation indicates that crack pattern and orientation were key contributors to the model's decision-making process.

4.3.2 | Crack Width Sensitivity

Although crack width was not measured explicitly, the dataset contains cracks ranging from hairline cracks (≈ 0.1 – 0.3 mm) to visibly wider cracks (> 1 mm). To analyze width sensitivity, a post hoc stratified evaluation was conducted by grouping test images into three visually estimated width categories (thin, medium, and wide) based on pixel thickness relative to image resolution. The proposed model achieved detection accuracies of 85.4% for thin cracks, 88.9% for medium cracks, and 91.6% for wide cracks, demonstrating increasing confidence with larger crack widths while maintaining strong performance on fine cracks.

4.3.3 | Crack Length and Continuity

Crack length and continuity were implicitly captured through the global contextual embeddings of the CLIP vision transformer. Longer and more continuous cracks tend to produce stronger global texture disruptions, which are effectively represented in transformer-based embeddings. In an auxiliary analysis conducted on 150 manually grouped samples, images containing long or continuous cracks covering more than 25% of the image span achieved a recall of 1.00, whereas images with shorter or fragmented cracks achieved a recall of 0.93. These results indicate that the model remains robust even for small-scale and discontinuous defects.

4.3.4 | Crack Depth Considerations

Crack depth cannot be directly inferred from 2D RGB images without additional sensing modalities such as stereo vision or depth sensors. Consequently, crack depth was not directly analyzed in this study. Nevertheless, surface cues correlated with depth including shadowing effects, intensity variations, and edge sharpness are partially captured within the learned visual features. This limitation has been explicitly acknowledged and motivates future work involving multimodal data sources, such as RGB–D imagery or photogrammetric reconstruction.

4.4 | Quantitative Performance Comparison

Earlier studies on concrete crack detection primarily rely on CNN backbones or CNN–Transformer hybrids to improve contextual understanding.

In contrast, the proposed CLIP-based model adopts a vision transformer pretrained via multimodal contrastive learning, enabling it to encode globally consistent and texture-sensitive representations. Unlike CNNs, CLIP's self-attention mechanism models long-range spatial dependencies, which is particularly advantageous for detecting thin, elongated, and irregular crack patterns.

The proposed CLIP-based crack detection framework provides clear methodological, performance, and deployment advantages over conventional CNN- and hybrid-based approaches.

CNN-based models, such as ResNet50, rely on localized receptive fields, which limit their ability to effectively capture:

- Long and discontinuous cracks.
- Distributed surface irregularities.
- Nonlocal structural dependencies.

In contrast, the proposed method employs a ViT-based CLIP architecture that utilizes global self-attention to model long-range spatial relationships across the entire image. This architectural design enables the model to:

Detect elongated and fragmented cracks more reliably.

Capture global texture disruptions rather than isolated edge patterns.

Remain invariant to crack orientation and scale.

As a result, the proposed approach achieves perfect recall (1.00), ensuring that no cracks are missed during inspection. In the context of SHM, false negatives pose a significantly higher risk than false positives, as undetected cracks may lead to catastrophic structural failures. While prior models emphasize high precision, they often suffer from moderate recall, resulting in undetected defects.

The proposed CLIP-based model achieves:

Recall = 1.00,

F1 score = 0.93.

This guarantees comprehensive crack detection, making the framework particularly suitable for safety-critical inspection pipelines where conservative detection strategies are preferred.

From a deployment perspective, CNN-based approaches typically require full network fine-tuning, leading to increased training costs and a higher risk of overfitting. In contrast, the proposed framework:

Employs a frozen CLIP backbone.

Trains only a lightweight MLP classifier.

Requires significantly fewer trainable parameters.

Achieves inference latency below 12 ms per image.

These characteristics make the model highly suitable for real-time deployment on edge or resource-constrained platforms, outperforming heavier CNN-based alternatives.

Furthermore, while CNN-based methods depend heavily on dataset-specific tuning, the proposed framework demonstrates that:

Large-scale pretrained representations can be effectively transferred to crack detection tasks.

Only shallow adaptation is required.

Performance remains stable across diverse surface and environmental conditions.

Overall, this positions the proposed CLIP-based approach as a scalable, transferable, and robust solution for real-world SHM systems.

4.5 | Result Analysis

The experimental outcomes demonstrate that the proposed CLIP-based model exhibits strong and stable learning behavior throughout training. Both the training and validation losses showed a consistent downward trend across epochs, confirming the network's ability to minimize classification error without overfitting. The training loss decreased from 0.7026 in the first

epoch to 0.2606 in the final epoch, while the validation loss declined to 0.2501, reflecting effective feature transfer from the pretrained CLIP encoder to the domain-specific crack detection task. Correspondingly, training accuracy improved from 59.7% to 89.6%, and validation accuracy peaked at 94.1%, signifying robust convergence and generalization.

The F1 scores for both the training and validation sets remained consistently high (above 0.93) from mid-training onward, indicating a favorable balance between precision and recall. Notably, the validation recall sustained a perfect value of 1.0 across all epochs, demonstrating that the model successfully identified every cracked surface in the validation dataset. This behavior aligns with the objective of damage detection systems, where false negatives (missed cracks) are far more critical than false positives. The steady validation precision of approximately 0.88 further confirms that the model maintained a strong ability to differentiate between cracked and intact surfaces even when faced with unseen data.

In the final test evaluation, the model achieved an accuracy of 88.2% and an F1 score of 0.93, reinforcing its consistent performance beyond the training data. The perfect recall (1.0) on the test set again highlights the model's sensitivity and reliability in detecting structural damage, a crucial requirement in civil infrastructure safety applications. However, the specificity of 0.3333 suggests that the model produced several false positives—classifying some intact surfaces as cracked. While this slightly lowers precision (0.875), such bias toward sensitivity is often desirable in safety-critical domains, where missing a single crack could lead to catastrophic failure.

The ROC AUC value of 0.8571 and the MCC of 0.5401 further validate the discriminative power of the proposed classifier. AUC values above 0.85 generally indicate strong class separability, while an MCC greater than 0.5 confirms a substantial correlation between predicted and true labels even under class imbalance. Together, these metrics confirm that the proposed model maintains a good trade-off between sensitivity and reliability.

Qualitatively, the results suggest that the pretrained CLIP encoder effectively captured high-level semantic and texture-based cues from concrete surfaces, enabling the downstream MLP classifier to learn fine-grained distinctions between crack patterns and intact areas. The model's rapid convergence within 10 epochs underscores the efficiency of the transfer learning approach, minimizing training cost while preserving accuracy.

Despite the strong overall performance, the relatively low specificity indicates that additional refinement could be achieved by enhancing negative sample diversity or integrating multiscale feature fusion techniques.

Although the proposed model achieves perfect recall and a high F1-score, the comparatively low specificity indicates a tendency toward false-positive predictions, reflecting an oversensitive detection behavior. This trade-off is common in safety-critical SHM applications, where minimizing false negatives is prioritized. Nevertheless, several strategies could be explored to improve specificity without compromising recall. Threshold tuning on the classifier output probabilities could help balance sensitivity and specificity by adjusting the decision boundary according to application requirements. Additionally, cost-sensitive learning, where higher penalties are assigned to

TABLE 4 | Quantitative performance comparison on the SDNET2018 dataset.

Study/model	Backbone	Precision	Recall	F1 score	Key strength
ResNet50 [14]	CNN (ResNet50)	0.906	0.628	0.742	Low false positives
MultiScaleCrackNet [15]	CNN	~0.88	~0.86	~0.87	Multiwidth crack capture
CNN–Transformer Hybrid [21]	Hybrid	~0.89	~0.90	~0.89	Improved global context
Proposed CLIP-MLP	ViT (CLIP)	0.875	1.000	0.933	Zero missed cracks

false-positive errors during training, may encourage more conservative predictions on intact surfaces. Ensemble voting strategies that combine predictions from multiple lightweight classifiers or diverse pretrained backbones could further enhance robustness by reducing noise-driven false alarms. These approaches represent promising directions for achieving a better trade-off between detection sensitivity and reliability in future extensions of this work.

A quantitative performance comparison between the proposed method and existing state-of-the-art approaches on the SDNET2018 dataset is presented in Table 4.

In summary, the result analysis establishes that the proposed CLIP-based classifier delivers high accuracy, perfect recall, and excellent F1-scores, confirming its suitability for automated crack detection in real-world SHM systems. The model achieves a desirable balance between sensitivity and generalization, ensuring reliable early detection of surface anomalies while maintaining computational efficiency for deployment on resource-constrained inspection platforms.

5 | Future Works

Although the proposed CLIP-based model achieved strong accuracy and perfect recall, several aspects can be further explored to enhance its generalization and applicability. Future work may focus on the following directions:

Multiscale Feature Fusion: Integrating hierarchical features from multiple convolutional or transformer-based encoders (e.g., ResNet50, ConvNeXt, or Swin Transformer) could improve the detection of both micro and macro cracks under diverse scales and orientations.

Attention-based Fine-tuning: Incorporating self-attention or cross-modal attention layers could allow the model to dynamically focus on salient regions, improving interpretability and reducing false positives.

Data Augmentation and Domain Adaptation: Expanding the dataset with synthetic crack patterns or employing domain adaptation techniques could make the model more robust to variations in lighting, weathering, and material composition.

Real-time Deployment: Optimization for embedded systems and edge devices (e.g., NVIDIA Jetson or Raspberry Pi) using model quantization and pruning could enable real-time crack inspection in field environments.

Explainability and Visualization: Future studies may incorporate explainable AI (XAI) techniques such as Grad-CAM or SHAP to visualize feature importance and provide insights into the model’s decision-making process.

Semi-supervised and Active Learning: Future research may explore semisupervised and active learning frameworks to leverage large volumes of unlabeled infrastructure imagery, thereby reducing manual annotation costs while maintaining high detection performance in large-scale SHM deployments.

These extensions would strengthen the system’s robustness, interpretability, and scalability for practical adoption in large-scale infrastructure monitoring networks.

6 | Availability of Resources

The implementation code of the proposed CLIP-based crack detection framework, along with other resources of the study, will be made publicly available to support reproducibility and facilitate future research.

7 | Conclusion

This paper presented a CLIP-based architecture for automated concrete crack detection using transfer learning. By employing a pretrained CLIP visual encoder as a fixed feature extractor and a lightweight MLP classifier for downstream learning, the proposed model achieved high detection accuracy with minimal computational overhead. Experimental results on the SDNET2018 dataset demonstrated strong performance, achieving 88.2% test accuracy, a 0.93 F1 score, and perfect recall. The findings confirm that leveraging pretrained multimodal representations can significantly enhance feature quality for texture-based defect recognition while reducing training complexity.

The approach offers a robust and scalable solution for real-world SHM systems, capable of operating efficiently in resource-constrained environments. Future work will extend this framework through multiscale fusion and domain adaptation strategies to further improve specificity and real-time performance in diverse inspection scenarios.

Funding

No funding was received for this manuscript.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Kaggle-Structural Defects Network (SDNET) 2018 at <https://www.kaggle.com/datasets/aniruddhsharma/structural-defects-network-concrete-crack-images/data>.

References

1. R. E. Philip, A. D. Andrushia, A. Nammalvar, B. G. A. Gurupatham, and K. Roy, “A Comparative Study on Crack Detection in Concrete Walls

- Using Transfer Learning Techniques,” *Journal of Composites Science* 7, no. 4 (2023): 169, <https://doi.org/10.3390/jcs7040169>.
2. L. Dai, M.-D. Cui, and X.-X. Cheng, “Structural-Health-Monitoring-Oriented Finite Element Model for a Specially Shaped Steel Arch Bridge and Its Application,” *Mathematical and Computational Applications* 28, no. 2 (February 2023): 33, <https://doi.org/10.3390/mca28020033>.
3. S. Moradi, B. Duran, S. E. Azam, and M. Mofid, “Novel Physics-Informed Artificial Neural Network Architectures for System and Input Identification of Structural Dynamics PDEs,” *Buildings* 13, no. 3 (February 2023): 650, <https://doi.org/10.3390/buildings13030650>.
4. J. Bardiani, R. Faure Ragani, L. Pinello, A. Kefal, A. Manes, and C. Sbarufatti, “Shape Sensing and Damage Detection of Composite Pressure Vessels Using Inverse Finite Element Method Coupled With Physics-Based Strain Pre-Extrapolation,” *Thin-Walled Structures* 218 (2026): 113935, <https://doi.org/10.1016/j.tws.2025.113935>.
5. B. Duran, D. Emory, Y. E. Azam, and D. G. Linzell, “A Novel CNN Architecture for Robust Structural Damage Identification via Strain Measurements and Its Validation via Full-Scale Experiments,” *Measurement* 239 (2024): 115393, <https://doi.org/10.1016/j.measurement.2024.115393>.
6. N. Lu, J. Cui, W. Zeng, Y. Liu, and M. Beer, “Damage Detection of a Cable-Stayed Bridge Specimen Based on Unsupervised Subdomain Adaptation Transfer Learning,” *Engineering Structures* 346 (January 2026): 121660, <https://doi.org/10.1016/j.engstruct.2025.121660>.
7. Q. Ge, C. Li, and F. Yang, “Research on the Application of Deep Learning Algorithm in the Damage Detection of Steel Structures,” *IEEE Access* 13 (2025): 76732–76746, <https://doi.org/10.1109/ACCESS.2025.3560640>.
8. J. Moll, C. Kexel, J. Kathol, et al., “Guided Waves for Damage Detection in Complex Composite Structures: The Influence of Omega Stringer and Different Reference Damage Size,” *Applied Sciences* 10, no. 9 (April 2020): 3068, <https://doi.org/10.3390/app10093068>.
9. A. N. Hoshyar, M. Rashidi, Y. Yu, and B. Samali, “Proposed Machine Learning Techniques for Bridge Structural Health Monitoring: A Laboratory Study,” *Remote Sensing* 15, no. 8 (April 2023): <https://doi.org/10.3390/rs15081984>.
10. A. Sharma, S. Gupta, and R. Kumar, “SDNET2018: Annotated Image Dataset for Non-Contact Concrete Crack Detection,” *Data in Brief* 19 (2018): 1513–1516, <https://doi.org/10.1016/j.dib.2018.06.027>.
11. S. S. Zadeh, S. Aalipour Birgani, M. Khorshidi, and F. Kooban, “Concrete Surface Crack Detection With Convolutional-Based Deep Learning Models,” *International Journal of Novel Research in Civil Structural and Earth Sciences* 10, no. 3 (2023): 25–35, <https://doi.org/10.5281/zenodo.10061654>.
12. S. Matarneh, M. Al-Sharif, and A. Alharthi, “Evaluation and Optimisation of Pre-Trained CNN Models for Asphalt Pavement Crack Detection,” *Computer-Aided Civil and Infrastructure Engineering* 39, no. 6 (2024): 1–15.
13. S. Kamada, T. Ichimura, and I. Takashi, (2021), An Adaptive Structural Learning of Deep Belief Network for Image-Based Crack Detection in Concrete Structures Using SDNET2018, <https://doi.org/10.48550/arXiv.2110.12700>.
14. M. S. Ansary and A. N. Suha, “Structural Damage Detection in Civil Infrastructures,” *European Journal of Electrical Engineering and Computer Science* 9, no. 6 (2025): 7–11, <https://doi.org/10.24018/ejece.2025.9.6.758>.
15. N. S. Russel and A. Selvaraj, “MultiScaleCrackNet: A Parallel Multiscale Deep CNN Architecture for Concrete Crack Classification,” *Expert Systems With Applications* 249 (2024): 123658, <https://doi.org/10.1016/j.eswa.2023.123658>.
16. A. M. Mayya, A. Selvaraj, and N. S. Russel, “Enhance the Concrete Crack Classification Based on a Multi-Scale CNN Architecture,” *Sensors (Basel)* 24, no. 24 (2024): 8095, <https://doi.org/10.3390/s24248095>.
17. F. Ding, (2024), Crack Detection in Infrastructure Using Transfer Learning, Spatial Attention, and Genetic Algorithm Optimization, <https://doi.org/10.48550/arXiv.2411.17140>.
18. F. Elghaish, M. Alsharif, and A. Alharthi, “Multi-Level Optimisation of Feature Extraction Networks for Concrete Crack Detection,” *Computer-Aided Civil and Infrastructure Engineering* 40, no. 1 (2025): 1–14.
19. D. Chen, H. Li, and L. Zhang, “Unsupervised Dam Crack Image Segmentation Algorithm Based on CD-CycleGAN,” *Computer-Aided Civil and Infrastructure Engineering* 40, no. 2 (2025): 1–14.
20. W. Qayyum, R. Ehtisham, A. Bahrami, et al., “Predicting Characteristics of Cracks in Concrete Structure Using Convolutional Neural Network and Image Processing,” *Frontiers in Materials* 10 (2023): 1210543, <https://doi.org/10.3389/fmats.2023.1210543>.
21. D. P. Yadav, B. Sharma, S. Chauhan, and I. B. Dhaou, “Bridging Convolutional Neural Networks and Transformers for Efficient Crack Detection in Concrete Building Structures,” *Sensors (Basel)* 24, no. 13 (2024): 4257, <https://doi.org/10.3390/s24134257>.
22. M. Li, Y. Zhang, and Y. Zhang, “CNN-Transformer Hybrid Network for Concrete Dam Crack Detection,” *Computers & Electrical Engineering* 106 (2024): 107460, <https://doi.org/10.1016/j.compeleceng.2024.107460>.
23. X. Sun, Y. Zhang, and Y. Li, “Concrete Crack Classification Based on Fourier Image Enhancement and Convolutional Neural Networks,” *Computer-Aided Civil and Infrastructure Engineering* 39, no. 7 (2024): 1044–1058, <https://doi.org/10.1111/mice.12983>.
24. Z. Yu, L. Wang, and X. Zhang, “Automatic Crack Detection on Concrete and Asphalt Surfaces Using Semantic Segmentation Network With Hierarchical Transformer,” *Journal of Civil Structural Health Monitoring* 14, no. 1 (2024): 1–14, <https://doi.org/10.1177/13694332251345935>.
25. M. O. Macaulay and M. Shafiee, “Machine Learning Techniques for Robotic and Autonomous Inspection of Mechanical Systems and Civil Infrastructure,” *Autonomous Intelligent Systems* 2, no. 1 (April 2022): 8, <https://doi.org/10.1007/s43684-022-00025-3>.
26. R. Dai, Y. Zhang, and Y. Li, “Crack Detection in Civil Infrastructure Using Autonomous Robotic Systems,” *Journal of Field Robotics* 42, no. 3 (2025): 1–14, <https://doi.org/10.1002/rob.22060>.
27. “Structural Defects Network (SDNET) 2018,” <https://www.kaggle.com/datasets/aniruddhsharma/structural-defects-network-concrete-crack-images/data>.